

MIKE WONG

mikedwong@cs.princeton.edu $\diamond$ [michaeldwong.github.io](https://github.com/michaeldwong) · [Google Scholar](#) · [LinkedIn](#)

SUMMARY

Fifth-year CS PhD candidate at Princeton building efficient ML systems for real-world deployment. Research thesis: data-efficient inference — using domain-specific signals to filter, compress, and prioritize data before expensive model stages. Published multiple papers at top venues (NSDI, SIGCOMM, CoNEXT, ECML PKDD). Two Microsoft Research internships and three years at NSA. Graduating late 2026.

EDUCATION

Princeton University *Aug 2021 – Late 2026 (expected)*
Ph.D. in Computer Science. Advisor: Ravi Netravali. M.A. conferred Sep 2023.

New York University *Aug 2015 – Dec 2019*
B.A. in Computer Science, minor in Mathematics.

EXPERIENCE

Graduate Research Assistant, Princeton University *Aug 2021 – Present*
Advisor: Ravi Netravali

- Built **MadEye** (NSDI '24, first author), a camera-server system that adapts PTZ orientations in real time for live video analytics. Combined custom knowledge distillation with a spatial search algorithm; boosts accuracy **2.9–25.7%** at fixed resource cost or hits same accuracy with **2–3.7 $\times$** lower cost. Outperforms PTZ tracking baselines by 2–3.8 $\times$ and multi-armed bandits by 5.8 $\times$.
- Lead author on position paper rethinking the role of programmable NICs in serving complex AI pipelines (arXiv 2502.15712); proposes pushing pipeline orchestration into network fabric.
- Contributing author on detection-guided attention steering for vision-language models (in submission).

Research Intern, Cloud Systems Reliability Group, Microsoft Research *May 2025 – Aug 2025*

- Built capability-aware execution system for LLM web agents that predicts when full ReAct loops are unnecessary, shortcutting to direct action. Reduces end-to-end latency and inference cost on web automation benchmarks.
- First-author paper in preparation with Kevin Hsieh, Suman Nath, Ravi Netravali.

Research Intern, Networking Research Group, Microsoft Research *Jun 2022 – Aug 2022*

- Built **NetVigil** (NSDI '24, co-first author), an east-west data-center anomaly detection system using GNNs and graph contrastive learning over low-cost flow summaries instead of packet traces.
- Reduced monitoring cost **2.7–16.7 $\times$** on 16-VM deployment vs. packet-based detectors; improved AUC by 0.22 average (up to 0.62) over state-of-the-art baselines. Led to filed US patent and open-source benchmark (Yatesbury).

Computer Scientist, National Security Agency *Feb 2020 – Aug 2021*

- Contributed to **Ghidra**, NSA's open-source reverse-engineering platform widely used across industry and government.
- Built **Marvolo** (ECML PKDD '23, first author), a programmatic data augmentation engine for ML malware detection. Applies semantics-preserving binary transformations and auto-propagates labels; improved MalConv2 detection accuracy up to 5% and AUC up to 10% on Ember dataset, with **79 $\times$** speedup over naive binary rewriting.

Research Assistant, Rutgers University / NYU Courant *May 2019 – Jan 2021*
Advisors: Srinivas Narayana, Anirudh Sivaraman

- Co-developed **K2** (SIGCOMM '21), a program-synthesis-based eBPF compiler using stochastic search over a Markov chain of rewrites. Reduced program size **6–26%** and packet-processing latency **1.36–55.03%** vs. clang -O2/-O3 across 19 Linux/Cilium/Facebook benchmarks. Accelerated equivalence checking by **5 orders of magnitude**.
- Built **Druzhba** (CoNEXT '20, first author), a programmable-switch hardware simulator for compiler testing; co-developed **Chipmunk** (SIGCOMM '20), a synthesis-based switch compiler with a slicing technique that speeds synthesis **1–387×** ($51\times$ avg).

Computer Science Co-op, National Security Agency

Aug 2016 – Dec 2019

- Developed a multithreaded linear algebra library in C++ using GMP and OpenMP to support cryptographic research workloads.
- Built Python-based static analyzers to detect vulnerabilities in assembly code; reverse-engineered applications to identify exploitable conditions.

SELECTED PUBLICATIONS

Full list at michaeldwong.github.io. * denotes equal contribution.

1. **MadEye: Boosting Live Video Analytics Accuracy with Adaptive Camera Configurations.**
Mike Wong, Murali Ramanujam, Guha Balakrishnan, Ravi Netravali. USENIX NSDI 2024. [\[paper\]](#)
2. **NetVigil: Robust and Low-Cost Anomaly Detection for East-West Data Center Security.**
Kevin Hsieh*, Mike Wong*, Santiago Segarra, Sathiya Kumaran Mani, et al. USENIX NSDI 2024. [\[paper\]](#)
3. **Marvolo: Programmatic Data Augmentation for ML-Driven Malware Detection.**
Mike Wong, Edward Raff, James Holt, Ravi Netravali. ECML PKDD 2023. [\[paper\]](#)
4. **Synthesizing Safe and Efficient Kernel Extensions for Packet Processing.**
Qiongwen Xu, Michael D. Wong, Tanvi Wagle, Srinivas Narayana, Anirudh Sivaraman. ACM SIGCOMM 2021. [\[paper\]](#)
5. **Testing Compilers for Programmable Switches Through Switch Hardware Simulation.**
Michael D. Wong, Aatish Kishan Varma, Anirudh Sivaraman. ACM CoNEXT 2020. [\[paper\]](#)
6. **Switch Code Generation Using Program Synthesis.**
Xiangyu Gao, Taegyun Kim, Michael D. Wong, et al. ACM SIGCOMM 2020. [\[paper\]](#)
7. **Capability-Aware Execution for Efficient Web Agents.**
Mike Wong, Kevin Hsieh, Suman Nath, Ravi Netravali. *In preparation*.

TECHNICAL SKILLS

Languages	Python, C++, C, Rust, Java, JavaScript
ML / Systems	PyTorch, computer vision, knowledge distillation, GNNs, model serving
Networking / Low-level	eBPF, Verilog, Linux kernel, P4, programmable switches, packet processing
Tools	Ghidra, Git, Docker, Linux, L ^A T _E X

PATENTS

Detecting Network Anomalies Using Network Flow Data. Tsuwang Hsieh, Santiago Martin Segarra, Sathiya Kumaran Mani, Srikanth Kandula, Michael Dean Wong. US patent App. 18/367,775.

TEACHING & SERVICE

Instructor, Prison Teaching Initiative (teaching incarcerated students): MATH 020 Elementary Algebra (Fall 2025); MATH 101 Number Systems (Spring 2025).

Teaching Assistant, Princeton University: COS 316 (Principles of Computer System Design, Fall 2023); COS 561 (Advanced Computer Networks, Spring 2023).